Every 2x2 game, counted

An exhaustive computational tour of pure-strategy Nash equilibria

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Why I built this

Game theory textbooks prove things. They tell you a 2×2 game has at most so many equilibria, that the Prisoner's Dilemma has a dominant-strategy outcome, that some games have no pure equilibrium at all. The proofs are clean but abstract - you never see what the *whole space* of games actually looks like.

So I brute-forced it. I generated every possible 2×2 payoff matrix over a fixed integer range, checked each one for Nash equilibria, classified it, and measured the results. No sampling, no theory - just enumerate everything and count.

The main deliverable is an interactive tool where you can type in any payoff matrix and instantly see its equilibria and game type. This write-up is what the data looks like in aggregate.

The object: payoff matrices and Nash equilibria

A payoff matrix is the basic unit of two-player strategic-form game theory. Rows are Player 1's strategies, columns are Player 2's. Each cell holds two numbers - what each player gets when they pick that row/column combination.

```
Col 0 Col 1
Row 0 (3, 3) (0, 5)
Row 1 (5, 0) (1, 1) ← Nash equilibrium
```

A **pure-strategy Nash equilibrium** is a cell where neither player can do better by unilaterally switching. Above (the Prisoner's Dilemma), Row 1 / Col 1 is the only one: either player switching drops their payoff from 1 to 0.

The method

Enumerate every cell-value combination. For 2×2 matrices with payoffs in 0-5, each of the four cells holds a pair of values from {0..5}, so there are 6^2 = 36 possibilities per cell and **36^4 = 1,679,616 distinct matrices**. Each gets checked for equilibria and tagged with structural properties.

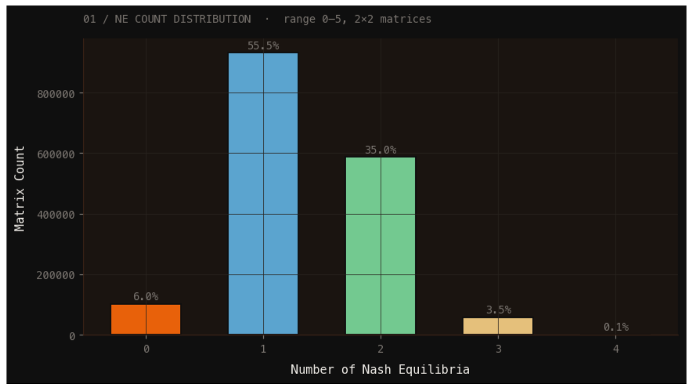
One thing worth stating up front: only the ranking matters

Pure-strategy equilibria depend on the *ordinal ranking* of payoffs, not the absolute numbers. A matrix with payoffs in 0-5 and one in 3-8 with the same relative ordering are strategically identical - shifting everything by a constant changes nothing.

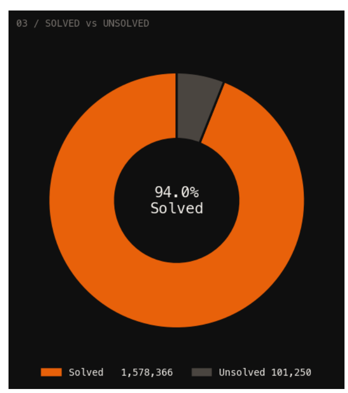
That means these integer ranges are really a **discrete sampling of the continuous payoff space**. The fractions I measure here (what % of games have an equilibrium, etc.) are empirical estimates of the probabilities studied analytically in the *random games* literature (Goldberg, Goldman & Newman; Rinott & Scarsini). I'm not proving anything new - I'm computing what those theoretical quantities actually look like at scale.

How many equilibria does a game have?

The headline number: in the 0-5 range, **94% of all 2×2 games have at least one pure equilibrium**. Most have exactly one.



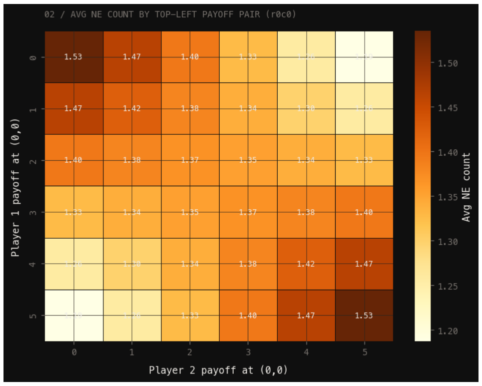
Equilibria	Matrices	Share
0	101,250	6.03%
1	931,500	55.46%
2	587,250	34.96%
3	58,320	3.47%
4	1,296	0.08%



The 6% with no pure equilibrium are exactly the games that need mixed strategies - I come back to those below.

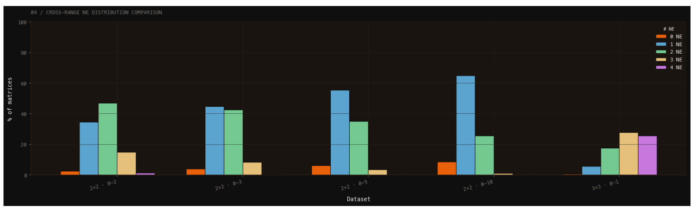
Does the payoff at one cell predict the equilibrium count?

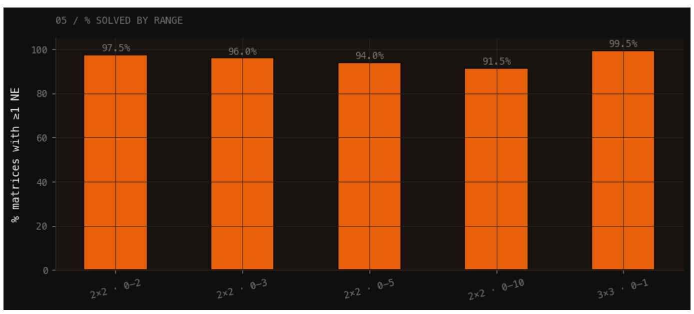
Grouping by the top-left cell's payoff pair and averaging the equilibrium count shows a clear gradient - the values in a single cell already shift the expected number of equilibria.



What happens as the range widens?

Here's the first genuinely interesting result. As you widen the payoff range, the fraction of games with a pure equilibrium goes **down**:





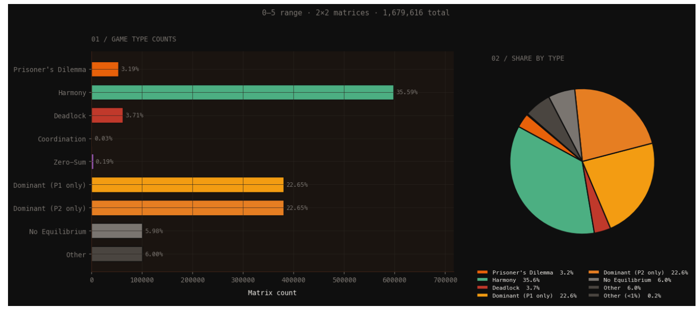
Range	Matrices	% with pure NE
0-2	6,561	97.53%
0-3	65,536	96.04%
0-5	1,679,616	93.97%
0-10	214,358,881	91.46%

The mechanism is ties. With a narrow range, lots of payoffs collide, and ties tend to *create* equilibria (a weakly-best response is still a best response). Widen the range, ties get rarer, and the equilibrium rate falls toward its continuous-payoff limit. That limit - the probability a random 2×2 game with continuous payoffs has a pure equilibrium - is exactly the kind of constant the random-games literature derives analytically. My 91.46% at 0-10, still trending down, is the discrete approximation creeping toward it.

(The 0-10 row has 214 million matrices. The raw file is ~8 GB, too big to load into the notebook, so its summary stats come from a pre-computed pass.)

What kind of game is it?

Counting equilibria is one thing; naming the game is another. A label like "Prisoner's Dilemma" carries a whole theorem - dominant strategies, a socially suboptimal outcome, cooperation that needs enforcement. I classify each matrix by its structural properties (dominant strategies, zero-sum, symmetry, Pareto efficiency of the equilibrium) and derive a named type.

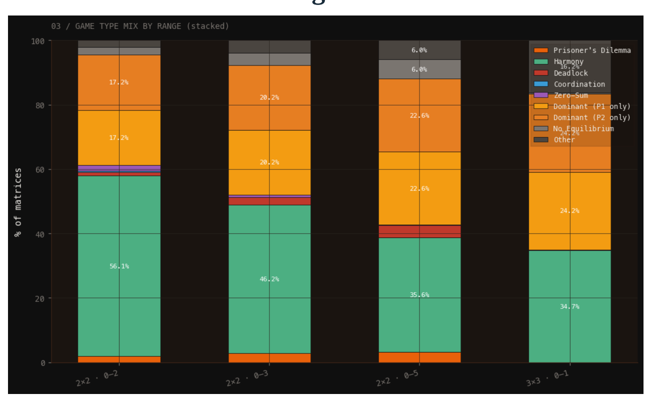


In the 0-5 range, the breakdown is dominated by benign games:

Game type	Share
Harmony	35.6%
Dominant (P1 only)	22.6%
Dominant (P2 only)	22.6%
Other	6.0%
No Equilibrium	6.0%
Deadlock	3.7%
Prisoner's Dilemma	3.2%
Zero-Sum	0.19%
Coordination	0.03%

The famous games are the rare ones. Prisoner's Dilemma is ~3%, true Coordination games are 3 in 10,000. Most random games are "Harmony" - the equilibrium is also the socially best outcome and there's no dilemma at all.

The mix shifts with the range



As the range widens, Harmony shrinks and the dominant-strategy games grow. A chi-squared test confirms the distributions genuinely differ across ranges:

```
chi^2 = 71,679.69
df = 24
```

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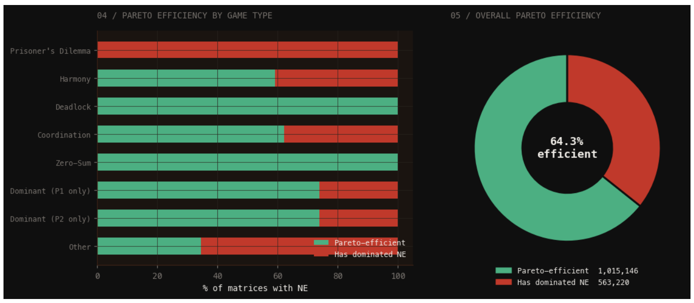
```
p ≈ 0
Cramér's V = 0.1089 (moderate effect)
```

With exhaustive data, the p-value is meaningless - every difference is real by construction since there's no sampling. The number that matters is **Cramér's V**, the effect size: 0.11 says the range *does* change the mix, but moderately, not dramatically.

Is the equilibrium any good?

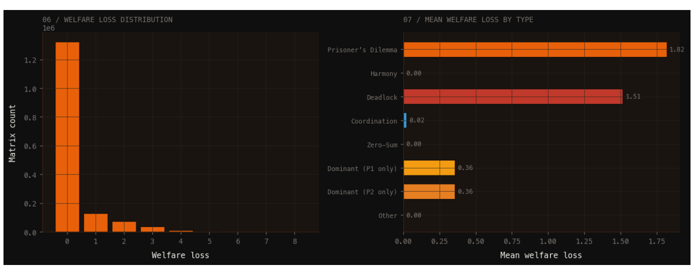
A Nash equilibrium is stable, not necessarily *good*. Two ways to measure that:

Pareto efficiency - is there another outcome that's better for both players? If so, the equilibrium leaves value on the table.



64.3% of games with an equilibrium have one that's Pareto-efficient. The rest - 35.7% - have an equilibrium that both players would happily trade away if they could coordinate. Prisoner's Dilemma is the extreme case: 0% efficient, by definition.

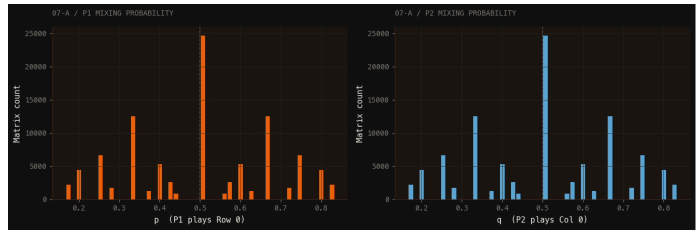
Welfare loss - quantify the gap. How much total payoff is lost at the equilibrium versus the best possible outcome?



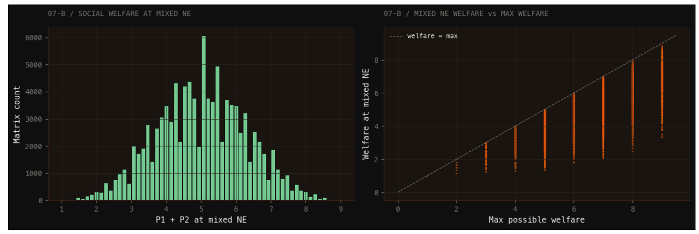
84% of games lose nothing - the equilibrium is socially optimal. The damage is concentrated in two game types: Prisoner's Dilemma (mean loss 1.82) and Deadlock (1.51). Everything else is near zero. The social-dilemma problem is real but rare.

The 6% with no pure equilibrium

For the 101,250 games with no pure equilibrium, Nash's theorem guarantees a mixed-strategy one exists - each player randomizes to keep the other indifferent. Every single one of them resolved to a valid mixed equilibrium; none were degenerate.



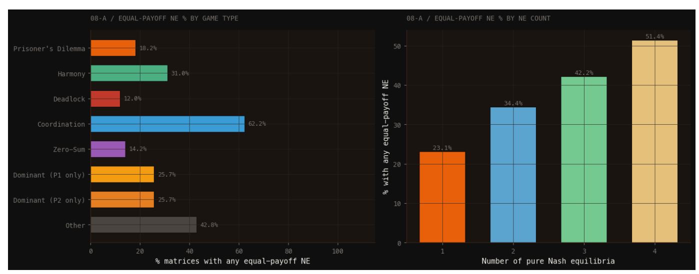
The mixing probabilities cluster hard at 0.5 (median p = median q = 0.5) - the symmetric games push both players to a coin flip - with secondary spikes at simple fractions like 1/3 and 2/3.



Mixed equilibria are costlier than pure ones: mean welfare loss 1.64, and only 0.7% hit the social optimum. Randomizing because you have to is rarely efficient.

Is the equilibrium fair?

Last question: when players reach an equilibrium, do they walk away with equal payoffs? "Stable" says nothing about "fair."



Across all games with an equilibrium, 28% have one where both players score equally. It varies enormously by type:

Game type	Equal-payoff NE
Coordination	62.2%
Other	42.8%
Harmony	31.0%
Dominant (either)	25.7%
Prisoner's Dilemma	18.2%
Zero-Sum	14.2%
Deadlock	12.0%

Coordination games are the fairest (players succeed or fail together); zero-sum and deadlock the least (one player's gain is the other's loss).

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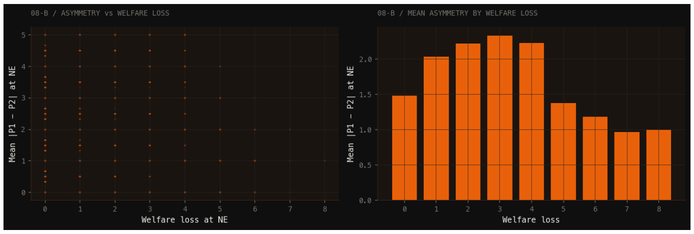
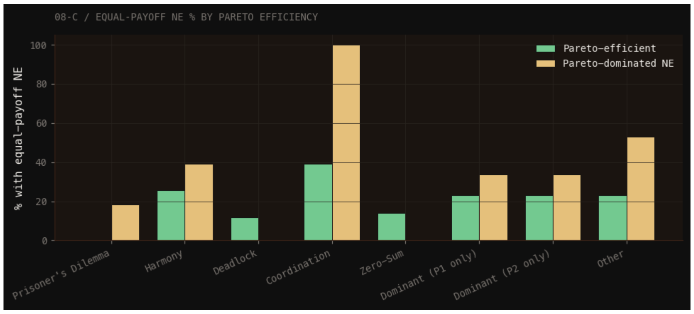


There's a weak positive correlation between payoff asymmetry and welfare loss (Pearson r = 0.18) - unfair equilibria tend to be slightly more wasteful - but it's not a strong link.



What this is and isn't

It isn't novel research. Equilibrium existence probabilities, the game-type taxonomy, the welfare results - all of this is known theoretically. The random-games literature derived the asymptotics decades ago.

The "correct" enumeration would be smaller. Since only ordinal ranking matters, the strategically distinct 2×2 games number in the hundreds, not millions. My integer datasets are massively redundant - every ordinal type appears thousands of times with different labels. Building the ordinal-permutation set is the obvious next step.

Coverage is uneven. 2×2 goes up to 0-10 (214M matrices); 3×3 only to 0-1 so far. The enriched 0-10 file is 24 GB, which is why the deeper analysis stops at 0-5.

What it *is* is a complete, explorable picture of a space that's usually only described in proofs - and a tool anyone can use to see where any given game sits in it. The empirical numbers line up with the theory, which is its own kind of confirmation.

Citation

Dunstan, A. L. (2026). Every 2x2 Game, Counted: An exhaustive computational tour of pure-strategy Nash equilibria. Game Theory Matrix Finder. https://github.com/Alex-Dunstan/game-theory-matrix-finder

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